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I rebuild manual processes as systems that run themselves.

Maritime workforce operations · Process improvement · Digitalization | Portfolio 2026

Nine projects from a maritime operation in Qatar: certificate compliance, crew travel, procurement, dispatch, service requests and workforce forecasting. All built on tools the company already owned, alongside the job of running crewing for 300+ seafarers.

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This portfolio describes how these systems work. It contains no screenshots, crew data, employee records or company material. Live walkthroughs available on request.

Impact at a glance

9	300+	25+	0
projects in this portfolio	seafarers supported	vessels, two sites	licence cost

How I build

Every system here began as something being done by hand, in an operation that could not stop while it was fixed. The method is the same each time.

01 **Watch the manual process first.** The people doing the work already know where it breaks. The job is to write it down precisely.

02 **Model the data before automating.** One source table beats five trackers. Most reporting problems are data-structure problems.

03 **Automate the repetition, never the judgement.** The system prepares the decision. A person still makes it.

04 **Deploy where people already work.** Excel, Outlook, Teams. No new platform to learn, no licence to buy, no adoption battle.

05 **Measure before and after.** If the saving cannot be stated as a number, the change was not worth making.

06 **Build it so someone else can run it.** Plain formulas, short flows, shared mailboxes. The system should keep working when I am on leave.

Tools: Microsoft 365 Power Automate (scheduled and event-driven flows) · Copilot Studio agents · applied language models for classification and summarisation · Excel modelling and VBA · SharePoint · Outlook and shared mailboxes · Teams · SAP, Workday and Crew Inspector as source systems

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Chapter One

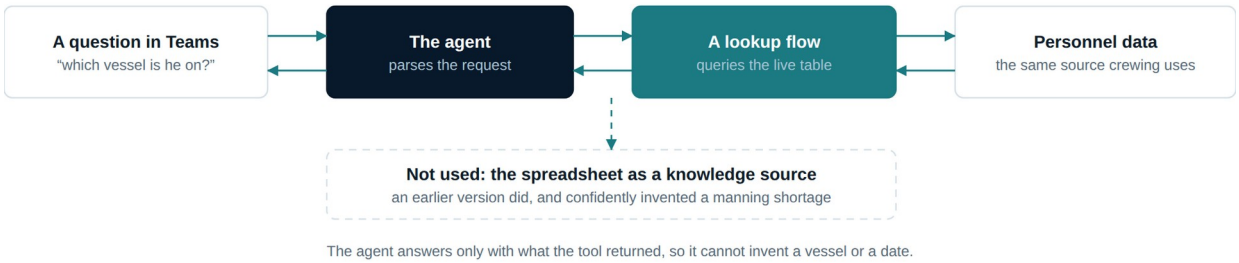
AI in operations

Two systems where a model does real work inside boundaries I set deliberately. Both are built so that being wrong is visible rather than quiet.

01 Bosun *my first working agent*

In service · 2026 · Agent

Routine crewing questions — which vessel is this crew member on, when is he due off, who is relieving him, what certificates is he still missing — all required opening the planner. Away from a laptop, or on leave, it meant calling whoever was in the office. The information existed; the access did not.



What it reads from, and what it deliberately does not.

What I built

An agent published to Microsoft Teams and usable from a phone, answering plain-English questions by crew name or vessel. It is backed by lookup flows rather than an indexed spreadsheet, so every answer comes from a live query against the same data crewing works from.

What it answers

- Where a crew member is right now, and which vessel he is allocated to
- When he is due off, and who his reliever is
- Who a given crew member is relieving
- Which certificates are still pending for crew arriving soon
- How many crew of each rank are missing on a vessel

How it works

1. The question is asked in ordinary language, by name or vessel rather than staff number.
2. The agent calls a lookup flow with the search term.
3. The flow reads the live personnel table and matches partial name or exact staff number.
4. A second lookup covers reliever data, which sits in a different table in the same workbook.
5. The agent answers only with what the tools returned.

Build log

The first attempt was deleted

An earlier agent used the workbook itself as a knowledge source. Asked about a vessel's engineer complement, it produced a confident, wrong answer. I deleted it rather than patch it, and rebuilt on tools only — an assistant that is occasionally wrong about manning is worse than none, because people stop checking.

The tool the agent could not see

The lookup flow existed and worked, but sitting in my own flows it never appeared in the agent's tool picker. Rather than keep hunting for the permission, I rebuilt the flow inside the agent itself and copied the expressions across from the original.

The field that lied about its own name

The flow kept failing on a null. The trigger input had been renamed to something readable, but its internal name never changed — the filter was matching against a label that did not exist. Referencing the internal name fixed it. Half a day, one line.

Working, and then extended

Partial-name search returns the single correct row. Once that held, I added reliever lookups and pending certificates, so it now answers the questions I was actually being asked rather than the one it was built for.

How I tried to break it

A crew member who does not exist Says so, rather than guessing at the nearest name.

A question about rank Answers from the tool, and only for the crew it actually returned.

A vessel question outside its tools Declines instead of improvising an answer.

"When am I off?" Resolved the asker, returned the correct date.

Result

- **Crewing questions answered from a phone, against live data**.
- **Grounded by design**: no spreadsheet indexing, so it cannot invent a vessel or a date.
- **It fails loudly**, which is the only reason it is safe to let other people use it.
- **Next: overdue counts by vessel and reliever availability**, then flow actions, controlled write-back and approvals.

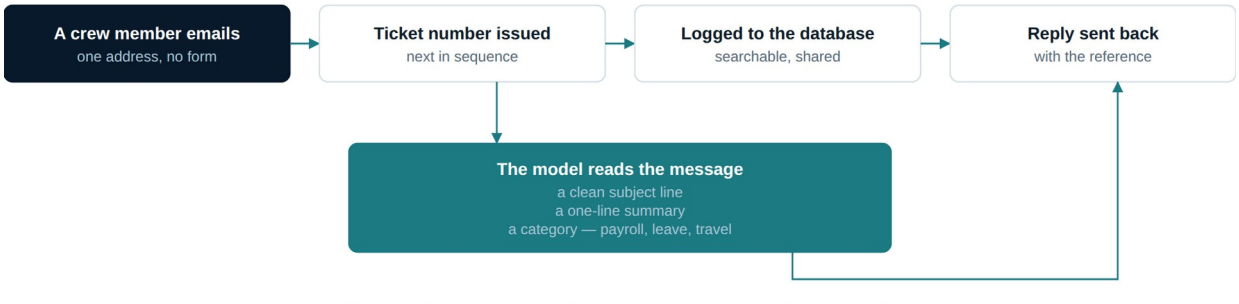
> This is the first agent I have built that anyone else can rely on. What made it work was not the model — it was refusing to let it read anything it might misread, and testing it on the questions where a wrong answer would have been expensive.

Skills demonstrated: Copilot Studio · Agent grounding · Power Automate tools · Failure-mode testing · Teams deployment

02 Helpline ticketing

Live · 2025 · Service desk

Crew emailed the helpline and every request was handled by hand: acknowledge it, rename the subject to staff number – name – request (or not, depending on who picked it up), search files for the sender's staff number and full name, work out the request, reply, then type it all into an Excel log. The same request looked different depending on who handled it.



Where the model sits, and what it is allowed to touch.

What I built

An email-to-ticket system: one address, automatic numbering, a searchable database, and a model that reads each message to produce a clean subject, a summary and a category before a person picks it up.

How it works

1. A crew member emails the shared address. No form, no training, no new tool.
2. The flow reads the last ticket number and issues the next one.
3. The request is written into the shared database.
4. The model produces a clean subject, a one-line summary, a category and a priority. If it fails, the ticket is still logged with safe defaults.
5. An automatic reply confirms receipt and gives the requester their reference.

Result

- **Manual steps per request cut from six to three.** The flow numbers the ticket, writes the subject, summary, category and priority, logs the row and acknowledges the sender. A person identifies the crew member, reads the summary and answers.
- **One format for every ticket**, whoever handles it.
- **Immediate confirmation with a reference**, which stops duplicate chasing.
- **A searchable history by category, crew, vessel and priority** — the raw material for fixing recurring causes.
- **Next:** identify the sender from their email address automatically. Tested against the log, it matches 220 of 290 past requests.

> The categorisation is triage, not judgement. It sorts and summarises so the person answering starts from something readable — it never decides an outcome, and a wrong category is visible on the ticket rather than buried.

Skills demonstrated: Power Automate · Applied language models · Service-desk design · Email-to-database integration

Chapter Two

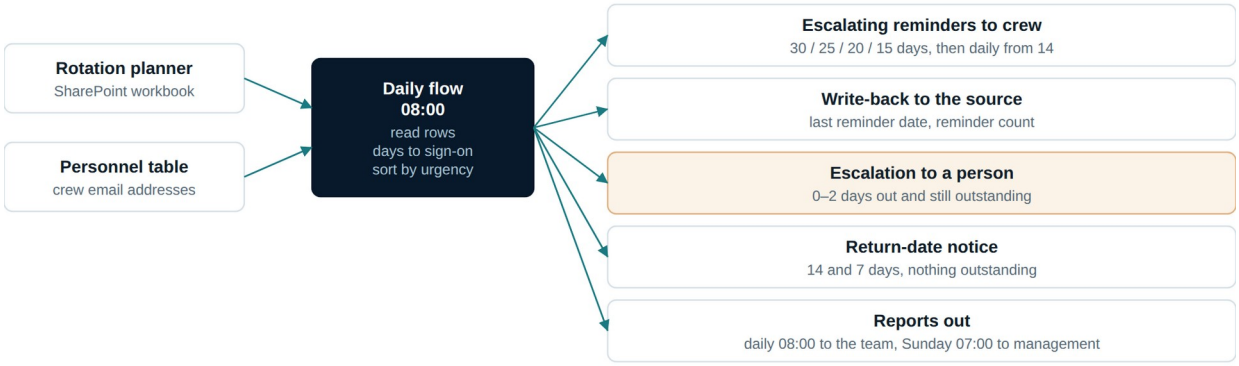
Process automation

Six processes that used to live in inboxes and memory. Each was written down, modelled, automated where the work was repetitive, and measured afterwards.

03 Certificate compliance chase

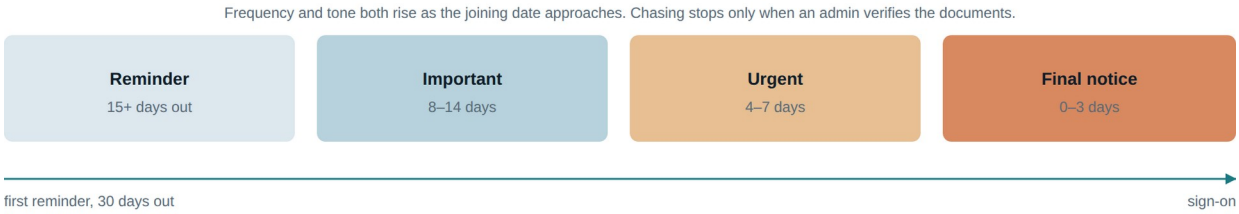
Live · 2026 · Compliance

A seafarer cannot join without valid certificates — safety training, medical, security awareness, watchkeeping endorsements, seaman book, passport. Chasing them was entirely manual: read the rotation planner, work out who is joining, check who is outstanding, write individual emails, and remember who was chased and when. Nothing was recorded, so nobody could prove a crew member had been told. Around 56 upcoming joiners had documents outstanding at any time, and a certificate missed until the joining week means a cancelled ticket, a short-manned vessel, or a crew member stopped at the gate.



A person verifies every certificate. The system flags a possible submission — it never marks anyone compliant.

The daily cycle, and the separate watcher that handles replies.



The escalation ladder.

What I built

A scheduled automation that reads the rotation planner every morning and runs the whole follow-up cycle unattended, plus a second, event-driven flow that watches the admin mailbox and puts any reply in front of a verifier within minutes.

How it works

1. At 08:00 the flow reads the rotation log and works out days to sign-on for every joiner.
2. It keeps the crew with certificates outstanding, sorts them most urgent first, and matches each to their email address.
3. Reminders go at 30, 25, 20 and 15 days, then daily from 14 days out until the documents arrive.
4. The source is updated with the last reminder date and count before anyone opens it that morning.
5. Crew within two days of joining go to the crewing manager instead of getting another automated email.
6. Crew with nothing outstanding get a return-date notice at 14 and 7 days, so joining dates are confirmed rather than assumed.

7. Replies are matched by sender address — crew rarely reply in thread — and posted to Teams with the certificate codes written out in full.

Result

- **Certificates now arrive 15–30 days before joining, not in the final week.** Before, crew joining within the week still had certificates outstanding, so they sat red on the planner and travel documents could not be sent. Within one week of go-live, almost every joiner 15–30 days out had submitted renewed certificates.
- **Follow-up runs whether or not anyone remembers it** — around 34 crew chased inside the 30-day window on a typical day.
- **Every reminder is evidenced**: date, count and status written back to the source.
- **Critical cases reach a person**, not another automated email.
- **Reminders send from shared mailboxes**, so the process survives its owner being on leave.

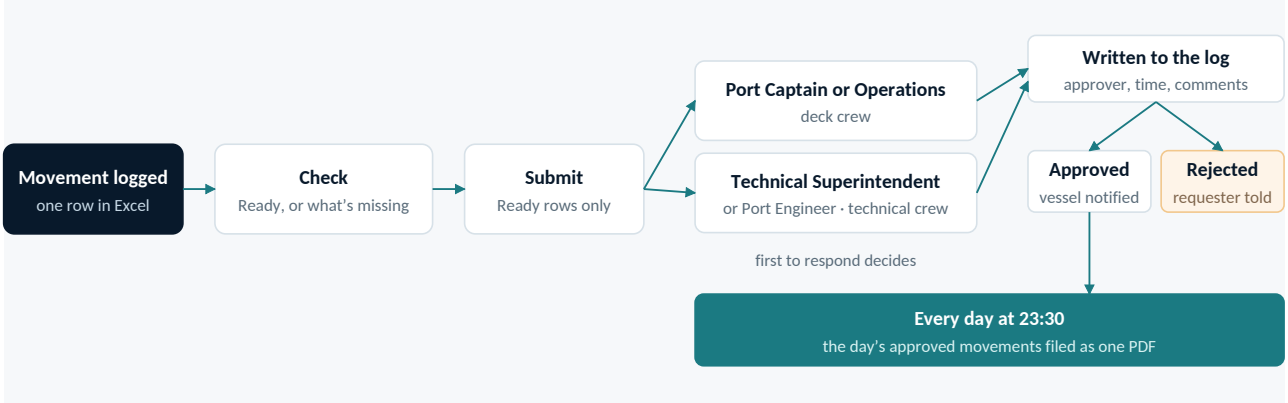
> Deliberately not automated: a person verifies every certificate. The system flags a possible submission; it never marks a crew member compliant. Crew occasionally send expired documents, so a reply alone can never be the proof.

Skills demonstrated: Power Automate · Scheduled and event-driven flows · Escalation design · MLC certification compliance

04 Crew movement approvals

Live · 2026 · Crew movements

Every crew movement — a relief, a shift between vessels, a temporary promotion or a familiarisation — went through the same hand-built chain: typed into an email, approved by reply, copied into a Word template with a pasted signature, converted to PDF, sent to the vessel or screenshotted to WhatsApp, saved in a folder, and keyed into the crewing dashboard by hand. A pasted signature proved nothing, and past approvals could only be checked by opening files one at a time.



One row in, one approval out, a record filed every night.

What I built

One Excel table as the log, and two flows on top of it. Lookups fill in names, ranks, department, approvers and the vessel contact, and a Check column says Ready or lists exactly what is missing. A submit flow handles each movement from request to notification; a scheduled flow writes the day's approved movements to one PDF. Routing lives in small settings tables, so changing an approver is a one-cell edit, not a flow change.

How it works

1. **Log.** Enter the vessel, crew staff numbers, type and dates. Lookups fill in the rest.
2. **Validate.** The Check column shows Ready, or exactly what is missing, including duplicates.
3. **Submit.** One button. Only rows that are Ready and not yet sent are picked up.
4. **Route.** Deck crew go to the Port Captain or Operations; technical crew to the Technical Superintendent or Port Engineer. The first to respond decides.
5. **Record.** Status, approver identity, a Qatar-time timestamp and comments are written back to the row.
6. **Notify.** Approved: a formal notice is emailed to the vessel. Rejected: the requester gets the reason.
7. **Archive.** At 23:30 each day, that day's approved movements are compiled into one PDF in the document library.

Result

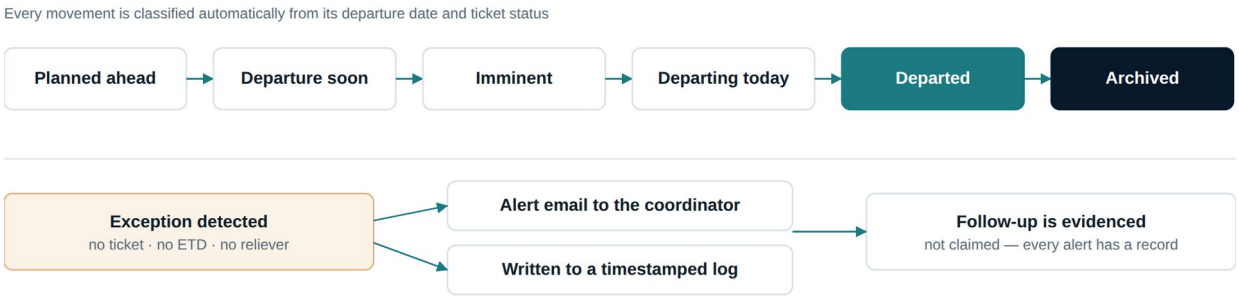
- **Manual steps per movement cut from 7 to 2** — log the row, press Submit.
- **Details typed once** — the approval, the email and the PDF all read from the same row.
- **Every approval carries a signed-in identity and a system timestamp** — no pasted signatures, and the flow's run history is a second audit trail.
- **One consolidated PDF a day** instead of one file per movement.
- **Next:** applying approved movements to the personnel roster automatically. Same-day escalation for unanswered approvals is designed and ready to switch on.

Skills demonstrated: Approval workflow design · Power Automate · Microsoft Approvals · Audit trail design

05 Departure ticket tracker

Live · 2025 · Crew travel

Crew travel lived in email threads between crewing, the travel agent and the vessels. Whether a ticket was booked, whether a reliever was assigned, and whether a departure was at risk could only be answered by searching an inbox. Bookings were duplicated, made too late, or quietly missed — and a missed departure means a vessel short-manned.



Status is derived from the data, and exceptions route themselves.

What I built

One register covering the whole lifecycle from planned to departed, where the status is derived rather than typed, and anything missing raises an alert against the person who owns it.

How it works

1. One row per movement: departure date, ticket status, visa type, vessel, destination, reliever.
2. Logic classifies every record automatically, from planned ahead through to departed.
3. Records with no ticket, no departure date or no reliever raise an alert.
4. Alerts go out as email and are written to a timestamped log.
5. Departed records archive themselves, so the working view only holds live movements.

Result

- **The whole travel pipeline visible in one place** — planned, booked, on hold and departed.
- **Lost and duplicated bookings eliminated**.
- **A departure with no reliever is caught days ahead**, while it is still a booking problem and not a manning failure.
- **Chasing replaced by an auditable log** of every alert sent.

Skills demonstrated: Process control · Exception alerting · Lifecycle modelling · Automated email

06 Procurement requisition workflow

Live · 2025 · Procurement

Requisitions arrived from vessels as free-form emails, routinely missing a GL code, WBS, part number, unit or quantity. Stores and purchasing spent their time chasing the gaps, approvals stalled, and between raising a request and receiving goods nobody could say where it stood.



Every request carries its own coding from the moment it is raised, so nothing waits on a missing field.

Validation sits before the queue, not after it.

What I built

A structured request with controlled fields and attachments, feeding a register that follows each line from submission to goods received. The validation is the point: an incomplete request never enters the queue.

How it works

1. The requestor completes a validated form covering vessel, category, priority, item, coding and department.
2. Submission issues a request ID and sends a fully coded email to stores.
3. Status moves through requisition issued, order issued, goods received, from stock, or rejected.
4. Dates, order numbers and material numbers are retained as the audit trail.

Result

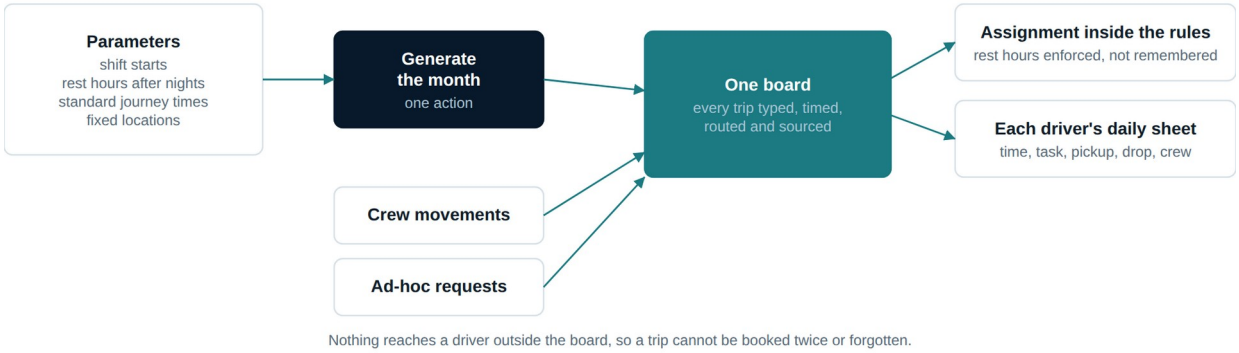
- **Approval cycle time reduced by around 60%**; requests no longer wait on missing information.
- **Incomplete requisitions stopped at source** — the form will not accept them.
- **A complete requisition-to-order trail with dates**, audit-ready without reconstruction.
- **One shared view across technical, stores and purchasing** instead of three private ones.

Skills demonstrated: Workflow design · Validation at the point of entry · Process standardisation · Audit trails

07 Driver scheduling engine

Live · 2025 · Logistics

Crew transport across two sites was dispatched by phone and memory. The same recurring runs — morning and evening crew change, food collection, airport transfers, the weekly out-of-town run — were retyped by hand every month. Ad-hoc requests arrived by call. Duplicate trips and missed pickups were usually discovered once crew were already waiting.



One board, whatever the source of the job.

What I built

A scheduling engine that generates a full month of recurring transport from a parameter sheet, absorbs everything else into the same board, and assigns drivers inside shift and rest rules.

How it works

1. Parameters hold shift starts, mandatory rest after night duty, standard journey times and fixed locations.
2. One action generates every recurring task for the month.
3. Crew movements and ad-hoc requests enter the same board, tagged by source.
4. Each driver gets a daily sheet with time, task, pickup, drop and the crew involved.

Result

- **A month of transport planning produced in one action** instead of line-by-line entry.
- **Duplicate dispatch eliminated** — every trip has one owner, one source, one status.
- **Rest-hour rules enforced by the schedule** rather than remembered by a dispatcher.
- **Spare capacity is visible**, so ad-hoc requests get a real answer.

Skills demonstrated: Scheduling engines · Resource allocation · Rest-hour compliance · Rules modelling

08 Night shift analysis

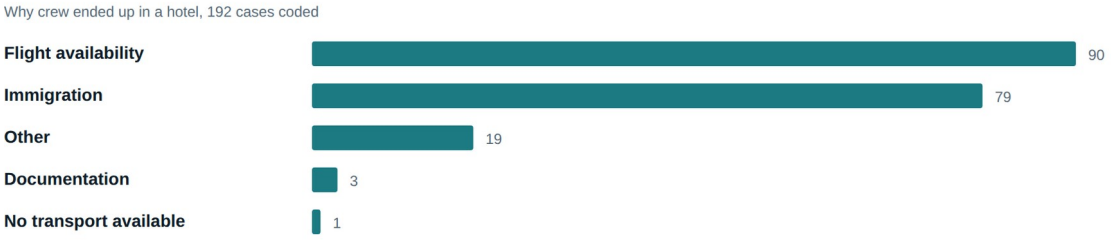
Delivered · 2024 · Cost analysis

Crew arriving on late flights were routinely booked into a Doha hotel overnight because no transport was rostered at that hour — around thirteen hotel nights a month, every month, plus the cost, the administration, and a lost day of the crew member's rotation. It had been accepted as unavoidable.



156 hotel nights across 2024, 36 across the ten months that followed.

The change, and whether it held.



Late flights and immigration were the causes of the delay. The night rotation removed the overnight stay that used to follow.

What was actually causing it.

What I built

Not software. Twenty-two months of hotel usage coded by root cause and tested against driver shift coverage, then a night rotation introduced to cover the window nobody was working.

How it works

1. Every hotel night logged by month, site and cause across 2024 and 2025.
2. Causes grouped and counted across 192 records.
3. Arrival patterns mapped against the driver roster to find the uncovered window.
4. A night rotation introduced to cover it, within rest-hour rules.
5. Usage tracked monthly afterwards to confirm the change held.

Result

- **From about thirteen hotel nights a month to three or four**, sustained across the following year.
- **A recurring cost removed from the crewing budget** — and evidenced, which made the next budget defensible.
- **Crew go straight to vessel or camp** instead of losing a night in Doha.
- **The remaining cases are genuine exceptions**, not routine.

Skills demonstrated: Root-cause analysis · Lean Six Sigma · Cost control · Data-driven rostering

Chapter Three

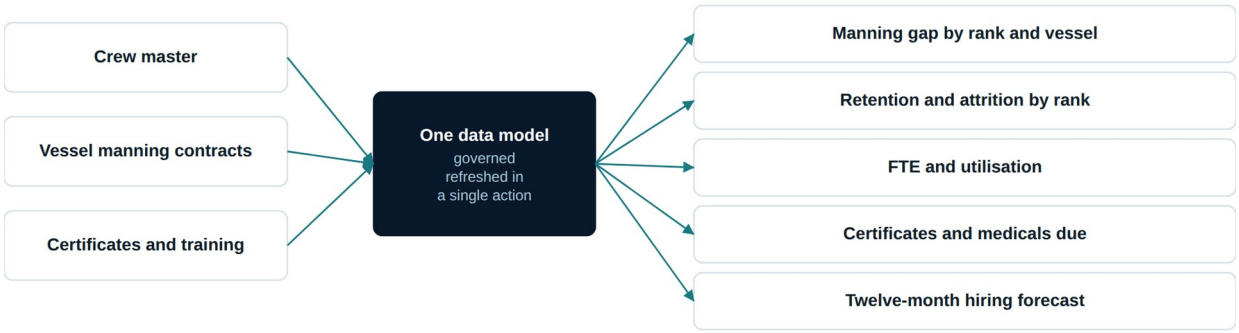
Crewing and workforce

The analytics layer underneath the day job: manning, retention, utilisation and a hiring forecast for a 300-strong seafaring workforce.

09 Crewing dashboard

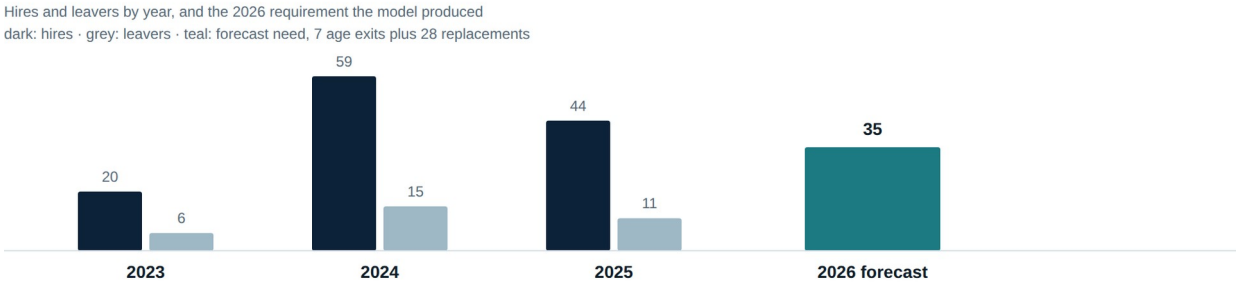
Live · 2024 · Workforce analytics

Manning status, certificate validity, joiner and leaver history and hiring needs lived in separate workbooks and inboxes. Building a monthly picture of 300+ crew across 25+ vessels took days of consolidation, and the answer was out of date by the time it was circulated. Nobody could say on demand which ranks were about to fall short, or how many people would need hiring next year.



Most reporting problems are data-structure problems. Fix the model once and every view follows.

One model, many views.



The forecast turns a yearly guess into a number recruitment can act on, broken down by rank.

What the forecast produced.

What I built

One data model over crew master data, manning contracts and certificate records. A single refresh rebuilds every view — manning gap, retention, utilisation, certificates due, and a rolling twelve-month hiring forecast.

How it works

1. Three source tables feed one governed model.
2. A refresh rebuilds all views and exception lists in a single action.
3. Exception logic flags overdue medicals, expiring certificates and crew reaching the age limit within twelve months.
4. The forecast converts planned exits plus historic attrition into a hiring requirement, by rank.

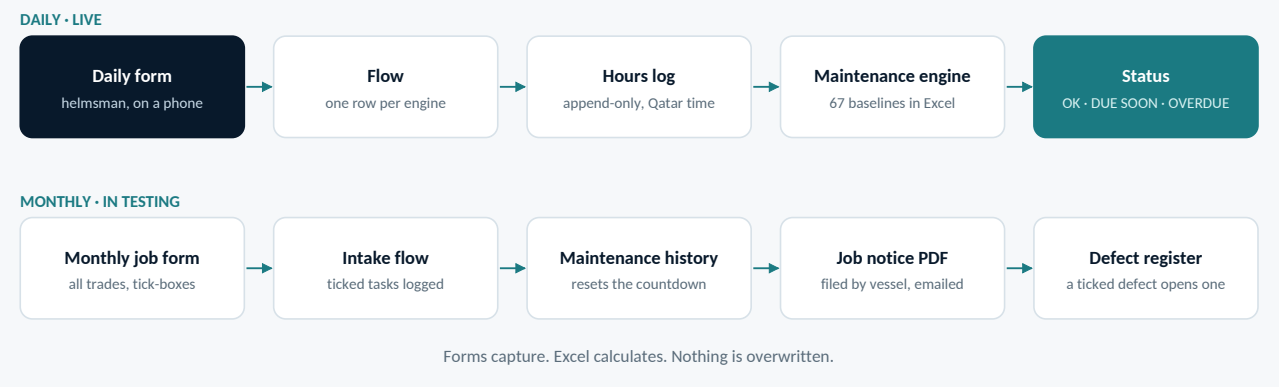
Result

- **Monthly reporting effort cut by around 60%** — consolidation replaced by one refresh.
- **Manning gaps surface by rank and vessel** before they become a failed crew change.
- **A 2026 hiring requirement of 35**: 7 age exits and 28 replacements, with the rank breakdown recruitment needs.
- **Compliance position audit-ready at any moment**, not reconstructed the week before an inspection.

Skills demonstrated: Data modelling · Workforce analytics · Manpower forecasting · Exception reporting

In progress: running hours and planned maintenance

Running hours were typed into a yearly workbook, often weeks old, and hours.minutes were added as decimals, so totals drifted. Maintenance due points lived in people's heads. Across nine small craft, daily capture is now live; the monthly job form and notices are in testing.



From a phone reading to a list of what is overdue.

- Daily capture is live: one phone form, one row per engine, hours.minutes converted to true decimals — fixing a long-running addition error.
- The first real data flagged 3 overdue lube-oil changes and 3 more due soon that nobody could see.
- In testing: monthly job form (96 questions cut to 4), PDF notices, defect register.
- Flows only append rows; the maintenance rules sit in Excel formulas the team can audit.

What this portfolio shows

These were not development projects. They were built alongside a full operational workload, without a budget, a development team or new software — using tools the company already owned, inside an operation that could not pause while they were introduced.

Each one started with a process being done by hand, was modelled before it was automated, was deployed where people already worked, and was measured afterwards. Where a language model does the work, its boundary is set deliberately: it sorts, summarises and surfaces, and a person still decides. That boundary is the part I would bring to a new operation, more than any individual system here.

Capabilities

- **Process automation** — scheduled and event-driven workflows, escalation paths, exception handling, write-back to source systems
- **Applied AI** — classification and summarisation in live workflows, agent design, grounding and hallucination control
- **Process analysis and improvement** — root-cause analysis, cost reduction, Lean Six Sigma applied to recurring spend
- **Data modelling and reporting** — one governed model behind many views, daily operational and weekly management reporting
- **Compliance systems** — MLC certification follow-up, evidenced reminders, authenticated approvals, audit trails
- **Workforce operations** — manning, rotation planning, manpower forecasting and budgeting for 300+ seafarers

Credentials

Lean Six Sigma Green Belt (2025) · NEBOSH IGC (2020) · Level 4 Diploma in Business Management · BSc Business Administration, Operations Management (graduating April 2027) · PMP in preparation

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References available on request.